

Ilankoon Pathiranage Pabasara Vikum Ilankoon

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Professional Summary

Electronic and Telecommunication Engineering undergraduate at General Sir John Kotelawala Defence University with a strong interest in **Embedded Systems, Artificial Intelligence, Machine Learning, Computer Vision, IoT and Wireless Communication**. Experienced in developing innovative engineering solutions through projects involving **AI-powered detection systems, embedded hardware integration, cloud-connected applications and long-range communication networks**. Proficient in Python, C++, Raspberry Pi, ESP32, OpenCV, YOLO and related technologies, with strong analytical, problem-solving and teamwork skills demonstrated through multidisciplinary engineering projects.

Education

General Sir John Kotelawala Defence University

BSc. (Hons) in Electronic and Telecommunication Engineering

Feb. 2024 - Present

CGPA: 3.02/4.00

University of Colombo School of Computing

BSc. in Information Technology (External)

Sep. 2024 – Present

St. Anne's College, Kurunegala

GCE Advanced Level (Physical Science Stream)

GCE Ordinary Level

Jan. 2013 – Feb. 2022

Relevant Coursework

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|------------------------|--------------------------|-------------------------------|
| • Signals and Systems | • Electromagnetics | • Embedded Systems |
| • Electronic Systems | • Electronic Circuits | • Digital Signal Processing |
| • Communication Theory | • Communication Networks | • Power Electronics |
| • Control Systems | • Digital System Design | • Image Processing and Vision |

Technical Skills

Programming

Python, C, C++, Arduino, Dart, JavaScript

AI/ML & Vision

Machine Learning, Deep Learning, OpenCV, YOLO, CNN, TensorFlow, Keras, MobileNetV2, Transfer Learning, Image Processing, Object Detection, Computer Vision, CVAT, Data Annotation

Embedded Systems

ESP32, ESP32-S3, ESP32-C3-Mini, STM32, Raspberry Pi, Arduino, Sensor Integration, Motor Control, Motor Drivers, PCB Design, Circuit Debugging, Power Systems

Robotics & Autonomy

Maze Solving Robots, Autonomous Navigation, Path Planning, Sensor Fusion, Obstacle Detection, PID Control, Real-Time Embedded Systems, PCB Designing, Robot Control Systems

Cloud & Software

Firebase, FastAPI, REST APIs, Flutter, React.js, Tailwind CSS, Git, GitHub, Mobile App Development, Web Application Development

Communication

LoRa, GSM, Wi-Fi, Bluetooth, UART, SPI, I2C, IoT Systems

Design Tools

EasyEDA, Proteus, Solidworks, LTSpice, MATLAB, VS Code, Arduino IDE, Microchip Studio, Tindercad

Research Experience

Elevison: A Real-Time AI-Powered Elephant Detection and Alert System for Railway Safety in Sri Lanka

- Researching an AI-based railway safety system for real-time elephant detection and early warning.
- Developing YOLOv8-based computer vision models for elephant detection in railway environments.
- Implementing edge AI deployment on Raspberry Pi using ONNX for real-time inference.
- Integrating Firebase, IoT communication, and Flutter applications for remote monitoring and alerts.
- Preparing a preprint manuscript on system design, implementation, and performance evaluation.

Keywords: Elephant Detection, YOLOv8, Railway Safety, Computer Vision, Edge AI, Firebase, Raspberry Pi, Flutter

Publication Status: Preprint Manuscript (Under Preparation)

Elevison | AI-Based Elephant Detection and Early Warning System



- Designed and developed a real-time AI-based railway safety system to detect elephants on or near railway tracks using computer vision and deep learning.
- Implemented YOLO-based object detection with OpenCV for real-time video frame analysis and wildlife detection.
- Built a web application for live monitoring, alert visualization and system status tracking.
- Developed a Flutter mobile application for real-time alerts and remote monitoring.
- Integrated Raspberry Pi edge computing, GSM communication and Firebase cloud services for real-time alerting and storage.

Technologies: Python, OpenCV, YOLO, CNN, Raspberry Pi, Firebase, GSM, IoT, Flutter, React.js

LankaMesh | LoRa-Based Disaster Communication System



- Designed and developed a decentralized emergency communication system for disaster environments without internet or cellular networks.
- Implemented LoRa-based long-range peer-to-peer communication for SOS alerts, messages and GPS data sharing.
- Designed embedded system architecture using ESP32-S3, LoRa RA-02 (SX1278) and GPS modules.
- Built a Flutter mobile application for emergency alert categorization and communication handling.
- Demonstrated scalable IoT-based disaster communication framework for resilient connectivity.

Technologies: ESP32-S3, LoRa RA-02, GPS, Flutter, IoT, Embedded Systems, Wireless Communication

AgroVision AI | Intelligent Tomato Leaf Disease Detection System



- Developed a full-stack AI system for tomato leaf disease detection using deep learning and computer vision.
- Built a MobileNetV2 transfer learning model for multi-class plant disease classification.
- Developed web (React.js) and mobile (Flutter) applications for image upload and prediction results.
- Implemented FastAPI backend for authentication, prediction and database management.
- Integrated Firebase cloud storage for image dataset handling and prediction history tracking.

Technologies: Python, TensorFlow, Keras, OpenCV, FastAPI, React.js, Flutter, Firebase, CNN, MobileNetV2

Digital Communication Simulator



- Developed a web-based simulation platform to model and analyze digital communication systems.
- Implemented modulation schemes including ASK, QPSK, 16-QAM and OFDM for system analysis.
- Simulated AWGN channel effects to evaluate signal degradation and noise impact.
- Built BER vs SNR performance evaluation tools for communication system comparison.
- Designed interactive signal visualization tools using Plotly and Matplotlib.

Technologies: Python, Flask, NumPy, SciPy, Plotly, Matplotlib, HTML, CSS, JavaScript, DSP

Mars Robot Competition | Autonomous Robotics System



- Designed and developed an autonomous robotic system for competition-based navigation and object handling tasks.
- Implemented sensor fusion using IR sensors, ultrasonic sensors, gyroscope and color sensors.
- Developed embedded control system using ESP32-S3 for real-time decision-making and motor control.
- Designed mechanical system using SolidWorks and 3D printing for structural fabrication.
- Implemented encoder-based motor control using TB6612FNG driver for precise movement control.

Technologies: ESP32-S3, C/C++, IR, Ultrasonic, MPU6050, TCS34725, SolidWorks, Embedded Systems

ROSCO'25 Competition Robot | Autonomous Line and Wall Following Robot



- Built an autonomous robot capable of line following, wall following and ramp navigation tasks.
- Implemented sensor fusion using VL53L0X ToF sensors, IR array and MPU6050 IMU.
- Developed embedded control system using ESP32-S3 and TB6612FNG motor driver.
- Designed chassis using laser-cut acrylic and custom 3D-printed mechanical parts.
- Integrated encoder-based feedback system for accurate motion control and stability.

Technologies: ESP32-S3, VL53L0X, IR Sensors, MPU6050, TB6612FNG, SolidWorks, Embedded Systems

Personal Finance Manager | Full-Stack Financial Tracking Web Application (Ongoing)

- Developing a full-stack personal finance management system to track income, expenses, budgets and financial insights.
- Implementing secure JWT-based authentication and RESTful APIs for user, expense, income and budget management.
- Building interactive dashboards with real-time financial analytics, spending trends and budget visualization.
- Integrating MongoDB database and Express.js backend for scalable financial data management.
- Developing a responsive React.js frontend with data visualization and secure client-server communication.

Technologies: React.js, Node.js, Express.js, MongoDB, JWT, REST APIs, Chart.js, bcrypt.js, HTML, CSS, JavaScript

FeelFill | Smart Blind-Friendly Measuring Cup (Ongoing)

- Developing a portable assistive device for blind users to measure liquid levels using non-contact capacitive sensing.
- Implementing external capacitive detection through cup walls for real-time liquid level monitoring without liquid contact.
- Using ESP32-C3 Mini for sensor processing, calibration, and multi-level detection.
- Designing vibration and buzzer feedback system with different alerts for fill levels.
- Building a compact clip-on device with rechargeable battery and low-power embedded design.

Technologies: ESP32-C3 Mini, Capacitive Sensing, Embedded Systems, PCB Design, IoT, Vibration Motor, Buzzer

Achievements

Champions	Target Sprint Shooting Competition - 2025
Precision Shooter Award	MSSC Shooting Championship - 2025
KDU Colours	Air Rifle - 2025

Certifications

AI/ML Engineer	2026, SLIIT; Machine learning, supervised and unsupervised learning, deep learning, neural networks, TensorFlow, Keras and model evaluation techniques.
PCB Designing Workshop	2025, IET KDU & Sri Lanka Institute of Robotics (SLIR); Schematic design, component placement, PCB layout design, design rule checking (DRC), Gerber file generation and PCB manufacturing workflow.
MATLAB Fundamentals	2024, MATLAB Training Program; Numerical computing, matrix operations, data visualization, scripting and engineering simulation techniques.

Leadership & Volunteering

Finance Team Lead: GENESIZ'26	May 2026 – Present
Finance Committee Member: ERIC KDU	Feb. 2026 – Present
Program Team Member: IEEE ComSoc KDU	June 2024 – Present
Design Committee Member: Mathematical Society KDU	Feb. 2025 – Feb. 2026
PR Team Member: ERIC KDU	Feb. 2025 – Feb. 2026

Professional Affiliations

IET Student Member: Institution of Engineering and Technology

References

Prof. T.L. Weerawardane Professor Dept. of Electronic and Telecom. Engineering General Sir John Kotelawala Defence University Email: tlw@kdu.ac.lk Tele: +94 77 444 1419	ENG. (Mrs.) P.N.Karunanayake Senior Lecturer Dept. of Electronic and Telecom. Engineering General Sir John Kotelawala Defence University Email: karunanayake@kdu.ac.lk Tele: +94 77 333 1189
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